

EXAM 7 – SPRING 2013

14. (2.25 points)

Given the following bond portfolio for an insurance company:

	<u>Years to Maturity</u>	<u>Bond Rating</u>	<u>Credit Risk Solvency Capital Requirement (\$000)</u>	<u>Bond Issuer's Current Annual Default Probability</u>
Bond 1	1	BBB	45	0.2%
Bond 2	3	BBB	157	0.2%

Assume that the best diversification can be approximated by correlation coefficients of 0.5.

a. (1.5 points)

Identify and briefly describe three sources of credit risk that may be associated with this bond portfolio.

b. (0.75 point)

Calculate the capital requirement due to credit risk for this bond portfolio assuming that both bonds are from the same issuer.

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Question 14 Sample Answer

Solution 1

- a) Direct Default: Both bonds have non-zero possibility to go into default and not paying the coupon/principal

Downgrade risk: role that bonds will downgrade and lose value in market and possibly increase the probability of default

Concentration risk: Only invest in BBB rated bond and no higher grade bonds. Also all bonds seem to be of lower rating type (no mention of industry/geography. But may still have concentration risk)

- b) Because they are from the same issuer I would assume the credit risk correlation is 1.00
In this case total capital required is= Cap Req+ Cap Req=45+157=202

Solution 2

- a) Default Risk: Risk that the bond issuer will default on the interest and/or principal payment of the bond

Downgrade: Risk that the bond will get a rating downgrade which will lower the value of the bond

Settlement Risk: risk related to the lag between the end data of the bond and the payment of capital

- b) $45 + 157 = 202 =$ Total Capital Requirement. The two values are added because they are 100% correlated

Examiner Comment

- a) The candidate was expected to identify and describe 3 credit risks.

Many candidates did not sufficiently describe the risk (Default risk is the risk of default) or described a different risk than identified. Some did not identify the risk. The question specifically asked for credit risk and financial risks were not allowed (interest rate risk, asset-liability mismatch, reinvestment risk, etc.)

Credit was given if the list included Key driver of credit risk (Credit quality, Maturity, Size of expected loss)

b) Bonds from the same issue should have an $\alpha = 0$ and the standard formula will simplify to the sum of the individual risks. And the answer should include $45+157=202$

Some credit was given for the formula, but candidates must show the final answer and how it was derived.